CARBONLEDGER

CARBON EMISSIONS REPORT

FZ Prestige Digital

1234 Market Street, San Francisco, CA 94103

Reporting Period: December 31, 2023 - December 30, 2024

Generated: July 20, 2025

GHG Protocol & ISO 14064-1 Compliant

EXECUTIVE SUMMARY

Unit Conversion Reference: All summary values in tonnes CO_2e (1 tonne = 1,000 kg). Detailed data tables display values in kg CO_2e for precision and audit trail compliance.

TOTAL EMISSIONS	DATA QUALITY	PRIMARY SOURCE	COVERAGE
166.4	**Excellent**	**Fuel**	**10**
tonnes CO_2e	rating	category	categories
Total organizational carbon footprint	Based on 36 verified data entries	Largest emission contributor	Emission sources tracked

Key Finding: Scope 1 emissions represent the dominant component of your carbon footprint at 54.9% of total emissions.

Data Integrity: This report is based on 36 verified data entries with excellent data quality standards, ensuring comprehensive coverage and audit readiness.

Audit Compliance: Report prepared in accordance with GHG Protocol Corporate Standard and ISO 14064-1:2018 requirements for third-party verification.

METHODOLOGY & STANDARDS

Reporting Framework

This carbon inventory has been prepared in strict accordance with:

GHG Protocol Corporate Accounting and Reporting Standard - Industry gold standard for corporate emissions reporting and verification

ISO 14064-1:2018 - International standard for greenhouse gas inventories and third-party verification processes

Operational Control Approach - Organizational boundary methodology as defined by GHG Protocol guidelines

IPCC AR5 Guidelines - Global warming potential values using 100-year timeframe for consistency

Calculation Methodology

All emissions calculations follow the fundamental equation: **Activity Data × Emission Factor = CO_2e Emissions**

Emission factors sourced from authoritative databases:

EPA Emission Factors Hub (2024) - Primary source for US-specific factors and regional grid electricity

DEFRA Greenhouse Gas Reporting Factors - International best practices and UK-specific methodologies

Regional Grid Electricity Factors - Location-specific energy emissions based on utility mix

IPCC Guidelines - Global warming potential values (AR5, 100-year timeframe)

Quality Assurance & Data Verification

Data Quality Assessment: Excellent

This assessment is based on data completeness, consistency, and verification procedures. The current score reflects exceptional data governance with comprehensive activity tracking and audit-ready documentation.

Verification Standards Applied:

- • Data consistency checks across all emission categories
- • Cross-validation of activity data against source documentation
- • Emission factor verification against authoritative databases
- • Completeness assessment for all material emission sources

EMISSIONS ANALYSIS

Scope Breakdown by Emissions Source

Scope 1 (Direct)

91.34

tonnes CO_2e (54.9%)

Scope 2 (Electricity)

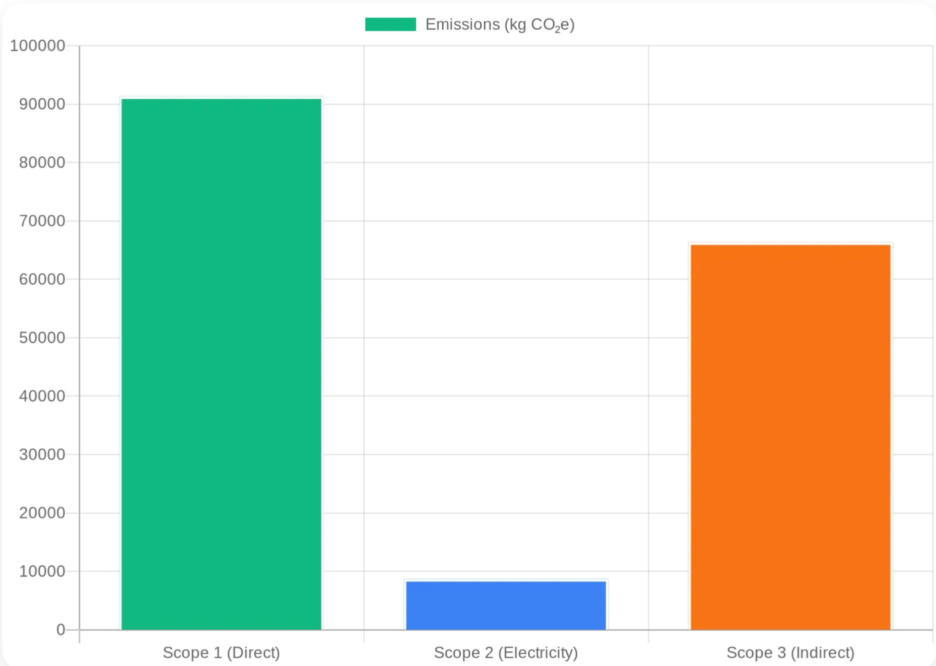
8.71

tonnes CO_2e (5.2%)

Scope 3 (Indirect)

66.34

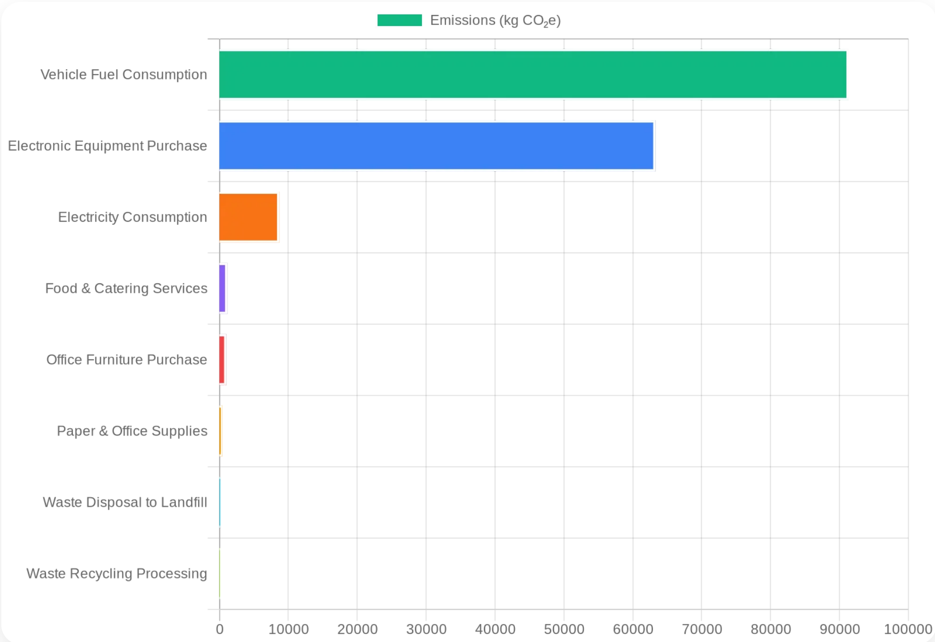
tonnes CO_2e (39.9%)



CATEGORY ANALYSIS

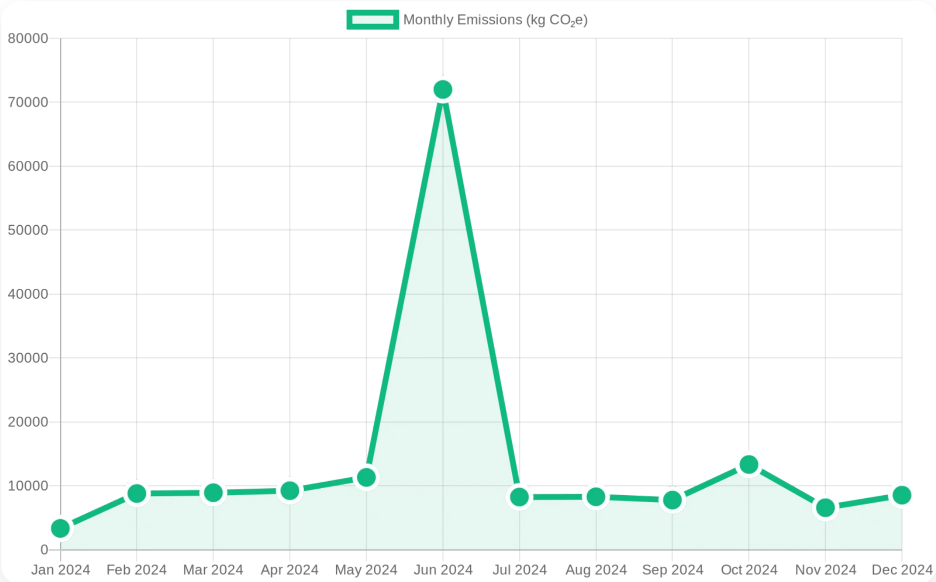
Top Emission Categories (Percentage of Total)

Fuel	54.9%	91,342.02 kg CO_2e
Purchased Goods Electronics	38.0%	63,300 kg CO_2e
Electricity	5.2%	8,706.98 kg CO_2e
Food Catering	0.7%	1,160 kg CO_2e
Purchased Goods Furniture	0.6%	1,020 kg CO_2e
Purchased Goods Paper	0.3%	426.88 kg CO_2e
Waste Landfill	0.1%	183.54 kg CO_2e
Waste Recycling	0.1%	118.86 kg CO_2e



Emissions (kg CO_2e)
Vehicle Fuel Consumption
Electronic Equipment Purchase
Electricity Consumption
Food & Catering Services
Office Furniture Purchase
Paper & Office Supplies
Waste Disposal to Landfill
Waste Recycling Processing
0
10000
20000
30000
40000
50000
60000
70000
80000
90000
100000

TREND ANALYSIS



Performance Metrics & Key Indicators

Average Monthly Emissions:	13.87 tonnes CO_2e
Peak Emissions Month:	72.00 tonnes CO_2e (2024-06)
Lowest Emissions Month:	3.34 tonnes CO_2e (2024-01)
Total Reporting Period:	166.39 tonnes CO_2e across 12 months
Emission Variability:	495.2% variation between peak and low months

STRATEGIC RECOMMENDATIONS

Immediate Priority Actions

Focus reduction efforts on Fuel as your primary emission source. This category represents the highest impact opportunity for carbon reduction initiatives and should be prioritized in your decarbonization strategy.

Strategic Focus Areas

Direct emissions control through operational efficiency programs, fuel switching initiatives, and process optimization.

Decarbonization Action Plan

→ **Short-term (0-6 months):** Implement energy efficiency measures, operational optimization, and quick-win reduction opportunities in high-impact categories

→ **Medium-term (6-18 months):** Develop comprehensive renewable energy procurement strategy, supplier engagement programs, and technology upgrade initiatives

→ **Long-term (18+ months):** Establish science-based targets aligned with 1.5°C pathway, comprehensive decarbonization roadmap, and net-zero transition plan

→ **Continuous Improvement:** Maintain current data quality excellence and expand verification procedures

DETAILED ACTIVITY DATA

Data Verification & Audit Trail: All activity data represents measured or estimated consumption using authoritative emission factors from EPA, DEFRA, and IPCC databases. Values are presented in source units with corresponding CO_2e calculations for full transparency, audit trail compliance, and third-party verification readiness.

DATE	SCOPE	CATEGORY	ACTIVITY DATA (UNITS)	CO_2E (KG)
01/31/2024	Scope 1	Fuel	1,232 liters	2,845.92
01/31/2024	Scope 2	Electricity	1,212 kWh	282.4
01/31/2024	Scope 3	Purchased Goods Paper	232 kg	213.44
02/29/2024	Scope 1	Fuel	3,233 liters	7,468.23
02/29/2024	Scope 2	Electricity	3,222 kWh	750.73
02/29/2024	Scope 3	Food Catering	232 meals	580
03/30/2024	Scope 1	Fuel	3,332 liters	7,696.92
03/30/2024	Scope 2	Electricity	4,343 kWh	1,011.92
03/30/2024	Scope 3	Purchased Goods Paper	232 kg	213.44
04/30/2024	Scope 1	Fuel	3,222 liters	7,442.82

DATE	SCOPE	CATEGORY	ACTIVITY DATA (UNITS)	CO_2E (KG)
04/30/2024	Scope 2	Electricity	3,344 kWh	779.15
04/30/2024	Scope 3	Purchased Goods Furniture	12 units	1,020
05/31/2024	Scope 1	Fuel	4,322 liters	9,983.82
05/31/2024	Scope 2	Electricity	3,244 kWh	755.85
05/31/2024	Scope 3	Food Catering	232 meals	580
06/30/2024	Scope 1	Fuel	3,432 liters	7,927.92
06/30/2024	Scope 2	Electricity	3,332 kWh	776.36
06/30/2024	Scope 3	Purchased Goods Electronics	211 units	63,300
07/31/2024	Scope 1	Fuel	3,221 liters	7,440.51
07/31/2024	Scope 2	Electricity	3,222 kWh	750.73
07/31/2024	Scope 3	Commuting	322 km	55.06
08/31/2024	Scope 1	Fuel	3,232 liters	7,465.92
08/31/2024	Scope 2	Electricity	3,222 kWh	750.73
08/31/2024	Scope 3	Waste Recycling	334 kg	70.14
09/30/2024	Scope 1	Fuel	3,222 liters	7,442.82

DATE	SCOPE	CATEGORY	ACTIVITY DATA (UNITS)	CO_2E (KG)
09/30/2024	Scope 2	Electricity	1,242 kWh	**289.39**
09/30/2024	Scope 3	Business Travel Car	232 km	**39.67**
10/31/2024	Scope 1	Fuel	5,432 liters	**12,547.92**
10/31/2024	Scope 2	Electricity	3,222 kWh	**750.73**
10/31/2024	Scope 3	Commuting	232 km	**39.67**
11/30/2024	Scope 1	Fuel	2,331 liters	**5,384.61**
11/30/2024	Scope 2	Electricity	4,332 kWh	**1,009.36**
11/30/2024	Scope 3	Waste Landfill	322 kg	**183.54**
12/31/2024	Scope 1	Fuel	3,331 liters	**7,694.61**
12/31/2024	Scope 2	Electricity	3,432 kWh	**799.66**
12/31/2024	Scope 3	Waste Recycling	232 kg	**48.72**

EMISSION FACTORS REFERENCE

Key Emission Factors Applied in Calculations

Energy & Fuels

Grid Electricity	0.233 kg CO_2e/kWh
Natural Gas	1.93 kg CO_2e/m^3
Gasoline	2.31 kg CO_2e/liter
Diesel Fuel	2.68 kg CO_2e/liter
Heating Oil	2.52 kg CO_2e/liter
Propane	1.51 kg CO_2e/liter

Transportation

Air Travel (Domestic)	0.255 kg CO_2e/pass-km
Business Travel (Car)	0.171 kg CO_2e/km
Rail Transport	0.041 kg CO_2e/pass-km
Freight (Truck)	0.062 kg CO_2e/t-km
Freight (Rail)	0.022 kg CO_2e/t-km
Freight (Air)	0.602 kg CO_2e/t-km

Purchased Goods & Services

Electronics	300.0 kg CO_2e/unit
Paper Products	0.92 kg CO_2e/kg
Food & Catering	2.5 kg CO_2e/meal
Office Furniture	85.0 kg CO_2e/unit

Waste & Water

Waste to Landfill	0.57 kg CO_2e/kg
Waste Recycling	0.21 kg CO_2e/kg
Water Supply	0.298 kg CO_2e/m^3
Wastewater Treatment	0.708 kg CO_2e/m^3

Authoritative Sources: *EPA Environmental Protection Agency (2024), DEFRA Department for Environment Food & Rural Affairs (2024), IPCC Intergovernmental Panel on Climate Change Guidelines (AR5)*